Step	Step Description	Agents Involved	Tool Calls and Parameters	Reason
1	User requests to find bungalows in Exeter	human, host_agent	Message from human to host_agent: 'Find Bungalows in Exeter'	The user initiates the query. host_agent receives and processes the request as the entry point for the system.
2	Plan the steps to solve the query	host_agent, planning_agent	send_message to planning_agent with task: 'Find Bungalows in Exeter. Provide the general steps to solve this query.'	planning_agent is chosen to decompose the query into actionable steps, as per its system prompt.
3	Planning agent outputs steps	planning_agent	Output: ["Find Exeter 1 area", "Search for bungalows in Exeter as many search results"]	Steps are based on the planning_agent's prompt, which instructs to first identify the area (Exeter) and then search for bungalows within it.
4	Find the area polygon for Exeter	host_agent, named_area_agent	send_message to named_area_agent: 'Find the area polygon for Exeter.'	named_area_agent is chosen because the prompt specifies it provides polygons for named areas such as cities.
5	Retrieve named area polygons	named_area_agent	call_os_ngd with: filters=None, bbox=None, polygon_or_point=True, filename='exeter_area_polygon'	named_area_agent uses the NGD tool to retrieve all named area polygons, as no filters are applied initially.
6	Filter to the specific Exeter polygon	host_agent, data_analysis_agent	send_message to data_analysis_agent: "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save as 'filtered_exeter_area_polygon'."	data_analysis_agent is chosen to filter the raw named area data to the specific polygon for Exeter, as multiple polygons may exist.
7	Generate metadata for artifact	data_analysis_agent	generate_metadata_for_artifacts with: artifact_names=['exeter_area_polygon']	Metadata is generated to understand the structure and columns of the artifact before filtering.
8	Filter Exeter polygon using code	data_analysis_agent, code_executor	code_executor with code to filter for Exeter using columns 'name1_text', 'description', 'descriptiongroup' and save as 'filtered_exeter_area_polygon'	The code uses multiple columns based on metadata to ensure accurate filtering for Exeter as a settlement.
9	Confirm filtered polygon artifact	data_analysis_agent	Output: Filtered to 2 polygon(s) for Exeter. Artifact: 'filtered_exeter_area_polygon'	The result confirms the filtering step and the artifact is ready for further spatial queries.
10	Search for bungalows within Exeter area	host_agent, buildings_agent	send_message to buildings_agent: "Search for bungalows within the area defined by the artifact 'filtered_exeter_area_polygon'."	buildings_agent is chosen as it handles building searches within a specified area (bbox).
11	Retrieve all buildings in Exeter area	buildings_agent	call_os_ngd with: filters=None, bbox='filtered_exeter_area_polygon', polygon_or_point=True, filename='all_buildings_in_filtered_exeter_area'	The NGD tool is called to retrieve all buildings within the filtered Exeter polygon, as 'bungalow' is not a direct filter.
12	Filter for bungalows using definition	host_agent, data_analysis_agent	send_message to data_analysis_agent: "From the artifact 'buildings_all_buildings_in_filtered_exeter_area', filter and extract all buildings that are bungalows. Use definition: residential (buildinguse_addresscount_residential > 0), only one address (buildinguse_addresscount_total = 1), single-storey (numberoffloors = 1). Save as 'bungalows_in_filtered_exeter_area'."	data_analysis_agent is chosen to apply the bungalow definition using the relevant columns, as per the buildings_agent's special scenarios and the system prompt.
13	Generate metadata for buildings artifact	data_analysis_agent	generate_metadata_for_artifacts with: artifact_names=['buildings_all_buildings_in_filtered_exeter_area']	Metadata is generated to confirm the presence and types of columns needed for filtering (residential count, address count, number of floors).
14	Filter bungalows using code	data_analysis_agent, code_executor	code_executor with code to filter for bungalows using the definition and save as 'bungalows_in_filtered_exeter_area'	The code uses the columns 'buildinguse_addresscount_residential', 'buildinguse_addresscount_total', and 'numberoffloors' as per the definition of a bungalow in the system prompt.
15	Confirm filtered bungalows artifact	data_analysis_agent	Output: Extracted 3,679 bungalows from 72,764 buildings in the filtered Exeter area. Artifact: 'bungalows_in_filtered_exeter_area'	The result confirms the filtering step and the artifact is ready for visualization or further analysis.
16	Plot the locations of bungalows	host_agent, plotting_agent	send_message to plotting_agent: "Plot the artifact 'bungalows_in_filtered_exeter_area' to show the locations of all bungalows in Exeter."	plotting_agent is chosen to visualize the spatial distribution of bungalows using a folium map, as per its system prompt.
17	Generate metadata for bungalows artifact	plotting_agent	generate_metadata_for_artifacts with: artifact_names=['bungalows_in_filtered_exeter_area']	Metadata is generated to confirm the structure and geometry of the artifact before plotting.
18	Plot bungalows using code	plotting_agent, code_executor	code_executor with code to plot all bungalows as polygons on a folium map, add markers for first 5, save as 'bungalows_in_exeter_map.html'	The code uses the geometry column and ensures CRS is EPSG:4326 for folium compatibility, as per plotting_agent's guidelines.
19	Confirm map artifact	plotting_agent	Output: Plotted 3,679 bungalows in Exeter. Artifact: 'bungalows_in_exeter_map', file: 'bungalows_in_exeter_map.html'	The result confirms the map was generated and saved, ready for user viewing.
20	Final output to user	host_agent	Output: I have found and mapped 3,679 bungalows in Exeter, using the definition of a bungalow as a single-storey residential building with only one address. The locations are shown on a map.	host_agent summarizes the results and provides the user with the outcome and the map artifact.

Overall Summary: - Databases used: Named area database (for Exeter polygon), Buildings database (for all buildings in Exeter), and plotting tools for visualization. - Filters and code: Filtering for Exeter used multiple columns ('name1_text', 'description', 'descriptiongroup') to ensure only settlement polygons for Exeter were selected. Bungalows were defined and filtered using 'buildinguse_addresscount_residential > 0', 'buildinguse_addresscount_total == 1', and 'numberoffloors == 1', as per the system prompt and metadata. - Outcome: 3,679 bungalows were found and mapped in Exeter. The locations were visualized on a folium map, with the artifact and map file provided to the user. - Reasoning: Each agent and tool was chosen based on their system prompts and capabilities. Parameters and columns were selected based on metadata and explicit instructions in the system prompts, ensuring accurate and relevant results at each step.